Question 1: [3 marks]

(a) [1 mark]

否定式

Given: $\neg E$
From $(I \wedge F) \rightarrow E$ and $\neg E$, apply Modus Tollens:
$\neg E \rightarrow \neg(I \wedge F) = \neg I \vee \neg F$
Conclusion: The person either did not have a valid ID or the fingerprint did not match (or both).

(b) [2 marks]

i. $\neg\forall x\ Fly(x)$
ii. Example: "There is a penguin in this region, and penguins cannot fly."

Question 2: [2 marks]

(a) [1 mark]

This rule can be read as: "If it is cold and someone is at home, then turn on the heating system."

In standard Boolean logic, this would require both inputs to be strictly True (1) to turn on the heating.

In an LNN, the $\otimes$ operator allows soft conjunction over continuous truth values. It supports partial inputs (like 0.4 or 0.9), yielding an intermediate activation that reflects uncertainty and permits gradient-based learning.

(b) [1 mark]
- Cold = 0.9

- AtHome = 0.4

Using soft-logic AND (e.g., Product-Sum), the output HeatingOn will reflect the multiplication of the two values, approx. 0.4 * 0.9 = 0.36.

Depending on the classification threshold, the system may or may not trigger the heating.

If the threshold is low (e.g., 0.3), heating will be turned on. If it's high (e.g., 0.7), it may stay off.

Question 3: [2 marks]

Entity and relation embeddings are vector representations of symbolic nodes (entities) and edges (relations) in a knowledge graph. Instead of treating each symbol independently, embedding methods map them into a continuous, low-dimensional space where semantic relationships can be captured numerically.

Common knowledge graph Inference tasks such as:
- Link prediction: Predicting missing links such as (h, r, ?) or (?, r, t)
- Relation prediction: Predicting missing links such as (h, ?, r)

Example:
Given embeddings of entities and relations, the model might discover that:
(Einstein, bornIn, ?) → Germany
even if this triple was not originally present in the KG.

Question 4 [2 marks]

The following strategies and collective behaviours were mentioned in class.

- Collective behaviours
    - E.g. a passing strategy
    - Suitable passing points extracted from interception prediction
    - Each passing point assigned a value based on its position
    - Plenty of room for optimization of such value assessment
- Positioning strategies
    - Choosing a formation on the field that yields good attack and defense opportunities
- Role-based strategies
    - Choose a role for each player based on game situation

Since all robots can access an overhead camera view in this league, they could all assess the game situation, attack/defense opportunities and suitable passing points on the same information. So, any of these behaviours/strategies appear feasible and useful.

Question 5: [3 marks]

(a) [2 marks] A random subset of the features would be sampled for any given tree in the random forest. Square-root of the total features was given as an example sample size, so sampling 15 features (√225) per tree would be a great choice; but anything substantially less than 225 could be OK. Typically, a large number of trees are produced (e.g. 2048), so we'd need to sample with replacement. So, *features would be selected as a random sample, with replacement, of size <<225.*

(b) [1 mark] Feature bagging is a solution to trees being highly-correlated, for instance due to a single strong feature being selected as the root of most trees (even on random subsets of the data). So, *feature bagging is considered to be a good idea because it makes the trees in the forest less correlated* and more able to complement one another as a ensemble.

Question 6: [3 marks]

The important thing about backward chaining is that the system tries to prove a specific goal. For this context, the goal of medical diagnosis is to decide what is likely to be wrong with the patient. Thus we would *start with one of the problems and find support for it in terms of symptoms*. We have three options, any of which are valid as 'an instance of backward chaining reasoning for a patient with runny nose.'

> Support 'common cold' – we've got all we need, as the patient has said they have runny nose
>
> Support 'measles' – we have runny nose, but we'd also have to find out if they have a distinctive rash
>
> Support 'allergies' – we have runny nose, but we'd also have to find out if they have history of allergies

Of course this is just a toy example. None of these rules would have 100% confidence. The doctor would need to consider the risk of being wrong (very high risk to send away a patient with measles letting them infect others) and the prevalence (thankfully, currently very low for measles).